

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q18-1078

Forest Lake 138 kV
December 01, 2019

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q18-1078. The Interconnection Customer proposes a 75.1 MW battery energy storage facility interconnecting at the Forest Lake 138 kV substation in the New Jersey area. The request was received on October 13, 2018. The study evaluated system intact and single contingency conditions using the regional planning case, including 418 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 4 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$1,342,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
RICLAND - TROY ENERGY LLC 500 kV Line	TREGO - SWAN VLY 500 kV	103.3	4,387	8.1
ASH STREET STA 223 SUBSTATION - TOMBSTONE 345 kV	TOMBSTONE - MARCUS HOOK ENERGY LP 500 kV	102.6	1,197	8.9
LAWRENCE - LAWRENCE SW 138 kV	MOTT SWITCH STA 345/138kV Transformer Bank	102.1	175	6.4
CORONADO 500/345kV Transformer Bank	MOJAVE SIHON SUBSTATION - CORONADO 345 kV	100.1	851	5.2

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
RICLAND - TROY ENERGY LLC 500 kV Line	Reconductor line	\$92,445,000	1% (\$922,000)
ASH STREET STA 223 SUBSTATION - TOMBSTONE 345 kV	Reconductor line	\$6,787,000	1% (\$78,000)
LAWRENCE - LAWRENCE SW 138 kV	Reconductor line	\$934,000	7% (\$69,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
CORONAO 500-345 kV Autotransformer	Replace transformer with higher-rated unit	\$23,828,000	1% (\$273,000)
	Total allocated to project		\$1,342,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.